

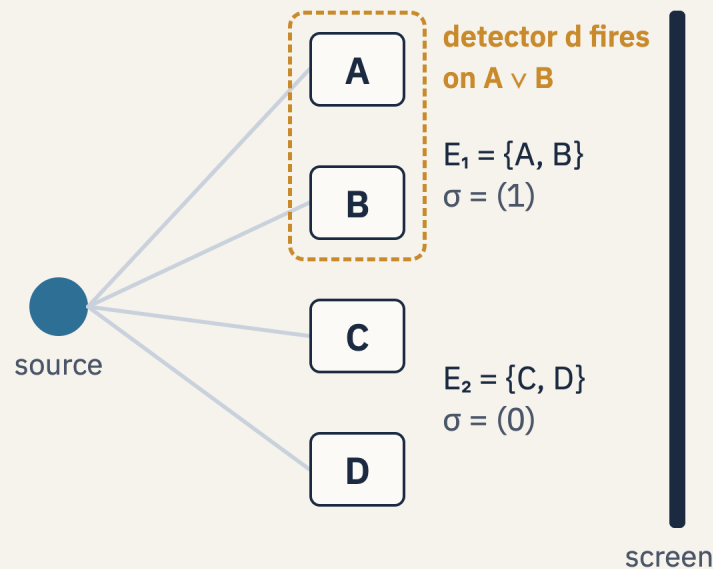
A FIFTEEN-MINUTE INTRODUCTION

Aethic Reasoning

Determinacy as a relation to an information state, and what the quantum measurement problem looks like once it is.

THE RESULT FIRST: THE POWERSET INVALIDATION THEOREM

The quantum–classical boundary is a partition, not a scale



$ T = 0$	$\emptyset$					
$ T = 1$	A	B	C	D		
$ T = 2$	AB	CD	AC	AD	BC	BD
$ T = 3$	ABC	ABD	ACD	BCD		
$ T = 4$	ABCD					

All 16 claims "the traversal is coherent over exactly T ", sorted by two rules and nothing else: no size, no mass, no observer.

✓ **valid: $T = \emptyset$ or $T =$ one whole class**
no refinement can expose a contradiction

✗ **rule 2: a proper part of a class**
a line drawn where no record ever could;
over-specification

✗ **rule 3: spans two classes**
some record separates them in every future;
dead now, not later

Move a record, move the boundary

NO DETECTOR



one class

valid: **ABCD** — fringes across all four

"But I can aim the beam at A." Yes: the collimator is a record, and {A} becomes a class of the new partition.

DETECTOR ON $A \vee B$



two classes

valid: **AB** **CD**

— fringes within each pair only

The record need not have been read. Its obtainability in every refinement kills ABCD now.

RECORD ERASED



one class again: no refinement can ever state the which-way record

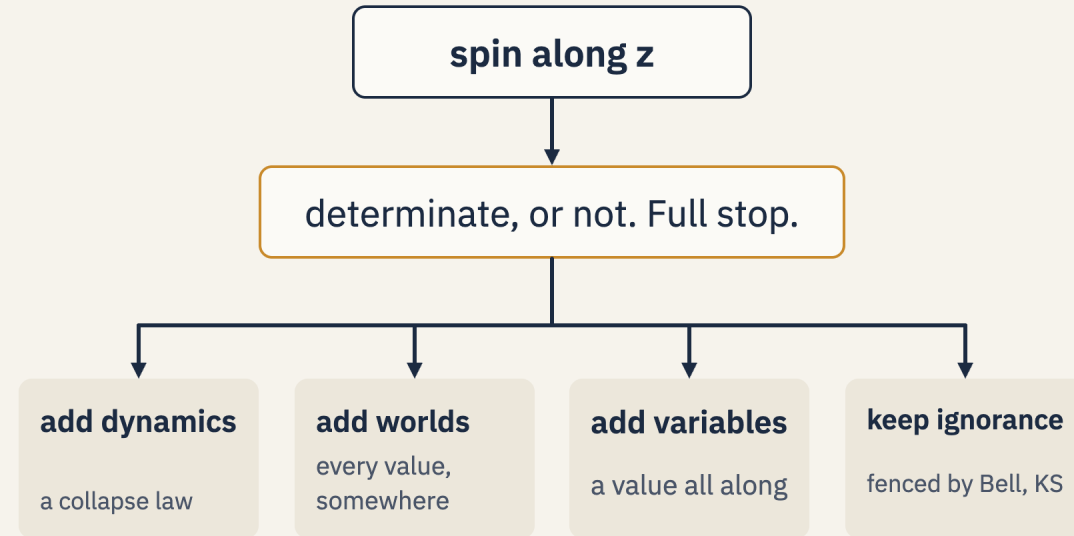
valid: **ABCD** — the eraser

Decided after traversal? Irrelevant: the classification is tense-free. A record that will never be readable was never a record.

Double slit, quantum eraser, delayed choice: **one theorem, re-run on three partitions.** So where do the two rules come from?

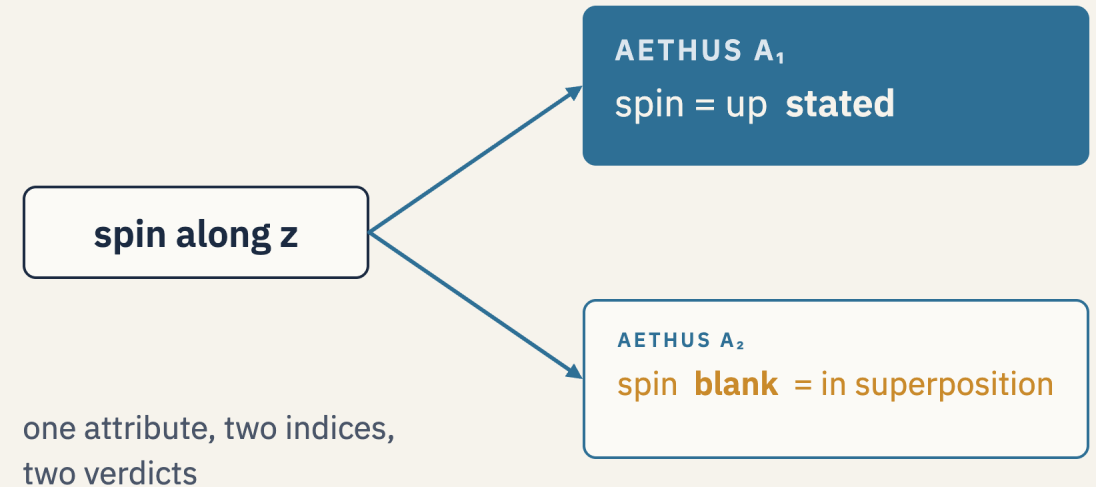
The measurement problem is a category mistake

STANDARD READING: DETERMINACY IS ONE-PLACE



Every answer keeps the notion and adds inventory.

AETHIC READING: DETERMINACY IS TWO-PLACE



Change the notion instead: a value is **relative to an information state**.

An Aethus: a closed partial ledger, with a tree beneath it

stated

a record fixes the value

blank

no record; open, not unknown

closed

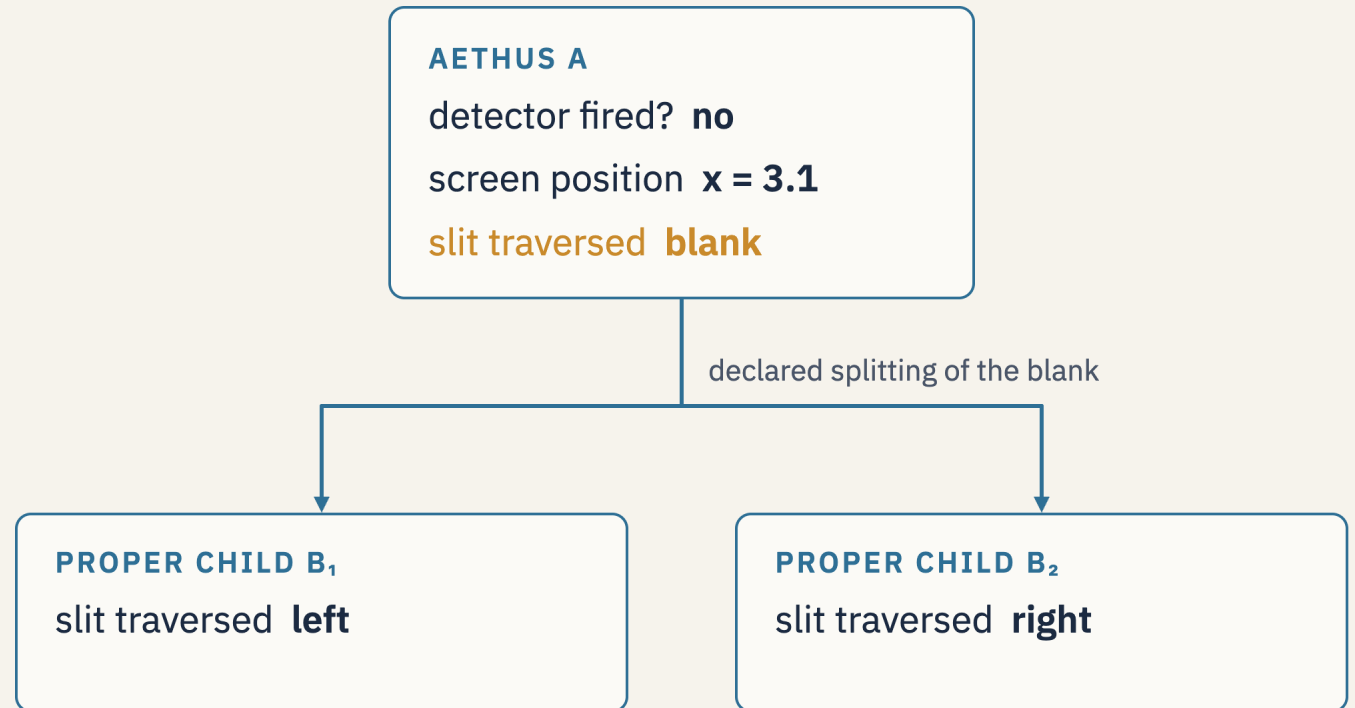
carries its own consequences under the standing laws

proper child

a blank resolved along a declared split

invalid (base)

stated content meets a declared joint invalidity



$B \leq A$, the Aethic tree: the order the third postulate quantifies over

Three postulates, stated exactly

FIRST

Any attribution of reality is a statement of relation to a particular Aethus.

SECOND

If an Aethus is valid, then any given attribute will be in Aethic superposition relative to it if and only if it is nonpresent to it.

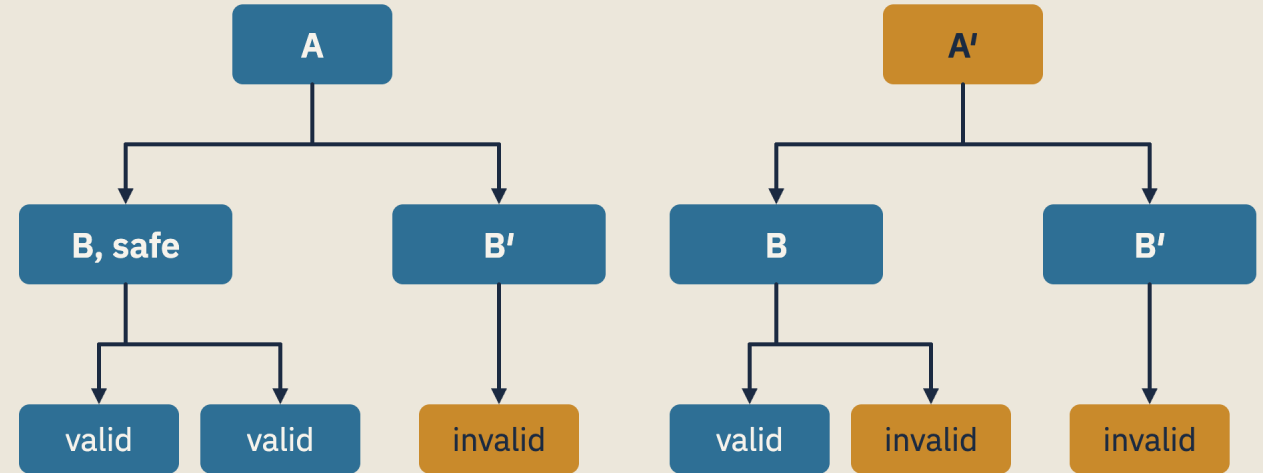
THIRD

If some Aethus is valid, then there exists a proper child Aethus to it for which every one of its own proper child Aethae are valid.

$$V[A] \Rightarrow \exists B \leq A \forall C \leq B : V[C]$$

An implication, not a biconditional: $V \subseteq vF$ always; $V = vF$ only on base generation.

$\exists$ A SAFE CHILD, $\forall$ ITS CHILDREN VALID



A valid. One safe child is enough.

A' invalid. No safe child: checkmate.

Jointness is primitive; exclusion is an achievement

AGREEING SUPERPOSITION



cross content undeclared $\rightarrow$ the alternatives are co-performed
 $\rightarrow$ **interference**

DISAGREEING SUPERPOSITION

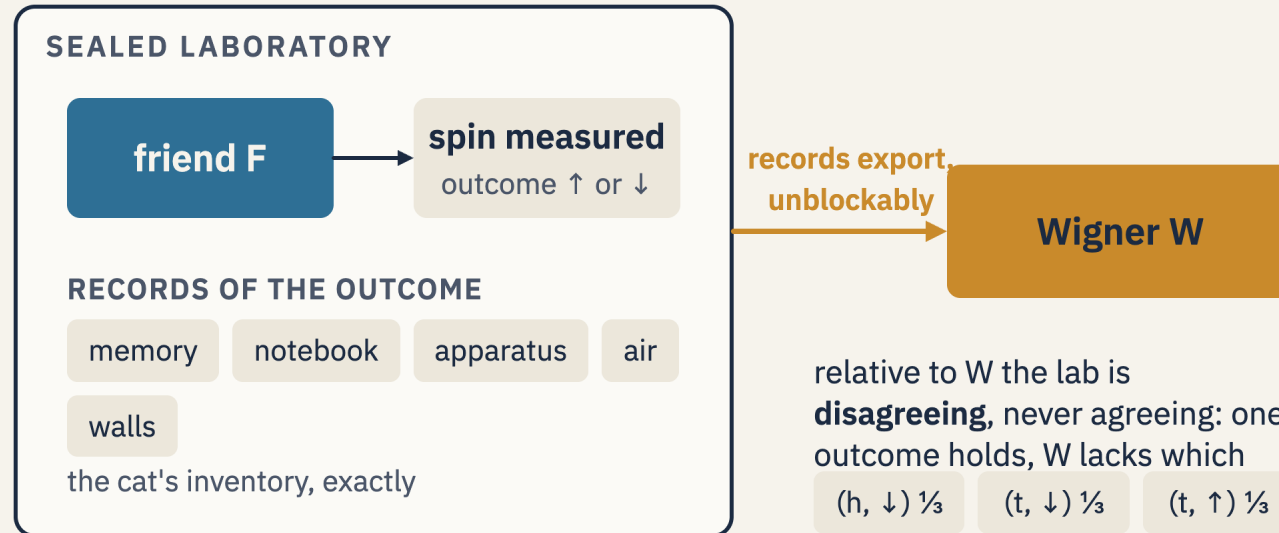


exclusions stated $\rightarrow$ weights over closed rooms $\rightarrow$ **the classical "or"**

The difference is written **in the alternatives**, not in the logic that composes them. A blank attribute is an agreeing superposition over its answer space.

Agreeing / disagreeing is qualitative and prior to any weight or phase; coherent / decoherent is quantitative, a property of a realization. No biconditional between them.

Wigner's friend: records forbid the cut, not scale



FRAUCHIGER-RENNER, JOINT OUTCOMES

	coherent lab	Aethic mixture
(ok_1, ok_2)	1/12	1/4
$(ok_1, fail_2)$	1/12	1/4
$(fail_1, ok_2)$	1/12	1/4
$(fail_1, fail_2)$	3/4	1/4

The certainty step ($ok_1 \Rightarrow$ spin up) needs the coherent assignment; the mixture never starts the chain. The (ok_1, ok_2) cell is the experiment.

The one divergence from universal unitarity: **an observer-grade system exports its records unblockably**. [Press here](#).

DEFLATION REDUCTIO

A blank is not an unknown

"BLANK" READ AS "UNKNOWN
VALUE" MEANS

D1 noncontextual hidden values

one value per attribute, whatever else
is measured

D2 verdict preservation

no classical verdict changes

D3 statistics = hidden- value frequencies

one measure μ ; measurement independence



FINE'S THEOREM

a joint distribution with
those marginals exists $\Leftrightarrow$
CHSH holds

$$|S| \leq 2$$



THE SINGLET, FROM THE KERNEL

$$|S| = 2\sqrt{2}$$

No lossless deflation exists.

The blank was constructive after all.

(a) Bell-violating sector

the agreeing regime; no single sample space

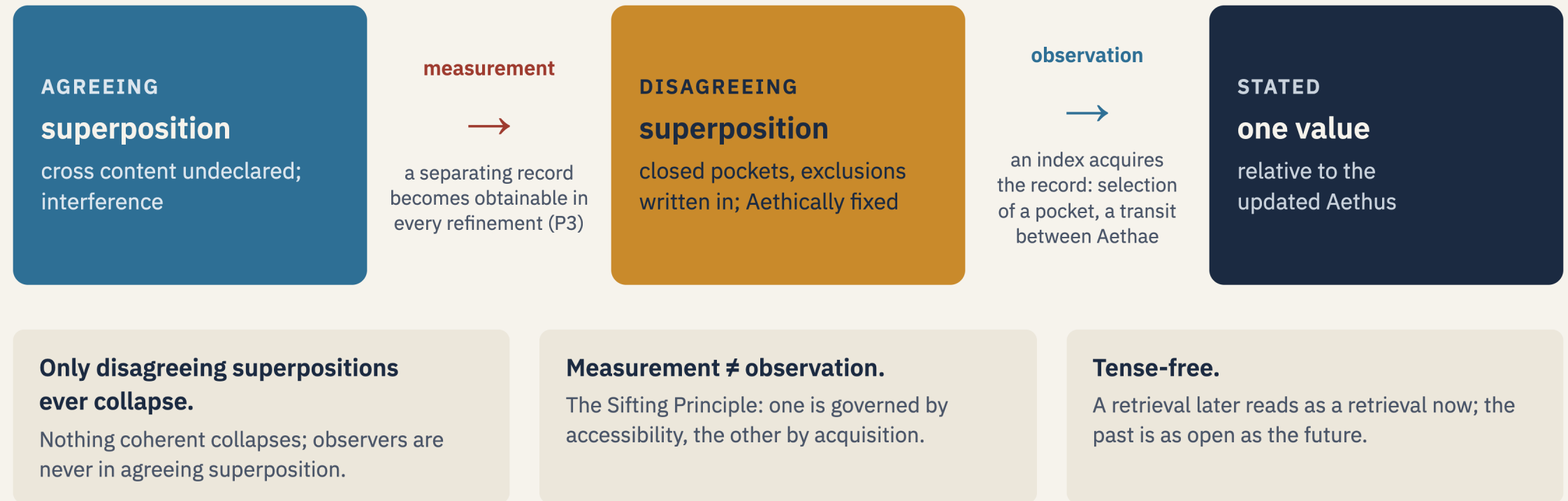
(b) Bell-bounded sector

the deflation exists: the classical filtration

The dilemma

deflate everywhere and fail, or halt exactly where the
statistics exceed the bounds

Collapse is a transit between indices, not an event



The same engine runs counterfactuals

"IF X WERE THE CASE..."

1 reopen
the target to a blank intermediate,
supports cleared



2 restate
the antecedent's value



3 re-prune
surviving pockets = the would-
consequences

AETHUS A

s stated

law: $s \rightarrow y$

$\{x, s, \neg z\}$ invalid

x, z blank

perturb by x

s rests on nothing cleared $\rightarrow$ kept $\rightarrow$ (x, s, y, z) $\rightarrow X \vdash \sim y$ and $X \vdash \sim z$

perturb by $x \wedge y$

y's support s is cleared $\rightarrow$ not restored $\rightarrow$ $(x, \neg s, y, \neg z)$ $(x, \neg s, y, z)$ (x, s, y, z)

$\neg z$ survives: $X \wedge y \not\vdash \sim z$

Cautious Monotony fails $\rightarrow$ not preferential $\rightarrow$ not Lewis, not Stalnaker, no smooth similarity ordering.

Machine-checked on the witness. The intermediate is the second postulate's blank; the pockets are what closure leaves.

Everything load-bearing, on one ledger

POSIT	GRADE	BUYS
1 the index	posited	two-place determinacy; the paradox set dissolved
2 the constructive blank	posited	the open past; the whole counterfactual semantics
3 primitive jointness of "or"	posited, from the cases	agreeing / disagreeing; the classical limit
4 $\exists \forall$ validity	derived in form, posited in class	checkmate; the Powerset Invalidation Theorem
5 one composition primitive	posited	the tree and its reduced form
6 zerosumfree semiring	design	validity separated from weight; nothing cancels a valid alternative
7 the kernel $\text{eiS}/\hbar$	the one quantitative posit	Dirac–von Neumann recovered on the finite path-kernel domain
8 the realization bridge	a principle	"invalid" $\Rightarrow$ "does not occur"

Not on the ledger, because derived: the projection postulate, no-signaling, the Born rule.

What would break it

A superposition that survives a record every refinement can read

refutes the third postulate

A cross-term surviving a record, or vanishing without one

refutes the Union Principle

The Bell statistics from a single sample space

refutes the constructive blank

An observer-grade laboratory coherently reversed: (ok_1 , ok_2) at 1/12, not 1/4

refutes the record premise, the one divergence from universal unitarity

Not a new mechanism. A changed notion of determinacy, with its bill itemized.